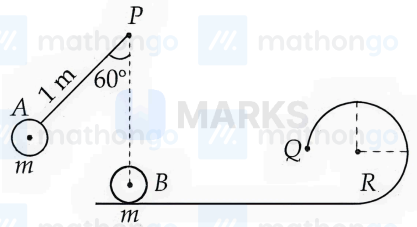
**Q1. JEE Main 2026 (21 January Shift 2)**

A body of mass 2 kg is moving along $x$-direction such that its displacement as function of time is given by $x(t) = \alpha t^2 + \beta t + \gamma$m, where $\alpha = 1$ m/s$^2$, $\beta = 1$ m/s and $\gamma = 1$ m. The work done on the body during the time interval $t = 2$ s to $t = 3$ s, is ____ J.

(1) 12 (2) 49 (3) 42 (4) 24

**Q2. JEE Main 2026 (23 January Shift 1)**

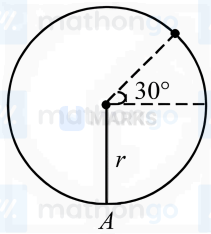
A small bob $A$ of mass $m$ is attached to a massless rigid rod of length 1 m pivoted at point $P$ and kept at an angle of $60^\circ$ with vertical as shown in figure. At distance of 1 m below point $P$, an identical bob $B$ is kept at rest on a smooth horizontal surface that extends to a circular track of radius $R$ as shown in figure. If bob $B$ just manages to complete the circular path of radius $R$ upto a point $Q$ after being hit elastically by bob $A$, then radius $R$ is ____ m.




(1) $\frac{1}{5}$ (2) $\frac{2-\sqrt{3}}{5}$ (3) $\frac{3}{5}$ (4) $\frac{2+\sqrt{3}}{5}$

**Q3. JEE Main 2026 (24 January Shift 2)**

In case of vertical circular motion of a particle by a thread of length $r$ if the tension in the thread is zero at an angle $30^\circ$ shown in figure, the velocity at the bottom point $(A)$ of the circular path is ($g$ = gravitational acceleration)




(1) $\sqrt{\frac{7}{2}gr}$ (2) $\sqrt{5gr}$ (3) $\sqrt{4gr}$ (4) $\sqrt{\frac{5}{2}gr}$

**Q4. JEE Main 2026 (22 January Shift 2)**

Given below are two statements :

**Statement I**: An object moves from position $r_1$ to position $r_2$ under a conservative force field $\vec{F}$. The work done by the force is $W = -\int_{r_1}^{r_2} \vec{F} \cdot \overrightarrow{dr}$.

**Statement II**: Any object moving from one location to another location can follow infinite number of paths. Therefore, the amount of work done by the object changes with the path it follows for a conservative force.

In the light of the above statements, choose the correct answer from the options given below :

(1) Both Statement I and Statement II are false

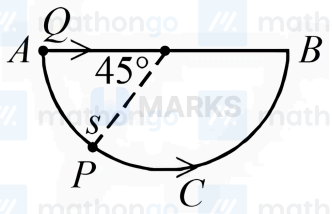
(2) Statement I is false but Statement II is true

(3) Statement I is true but Statement II is false

(4) Both Statement I and Statement II are true

**Q5. JEE Main 2026 (23 January Shift 2)**

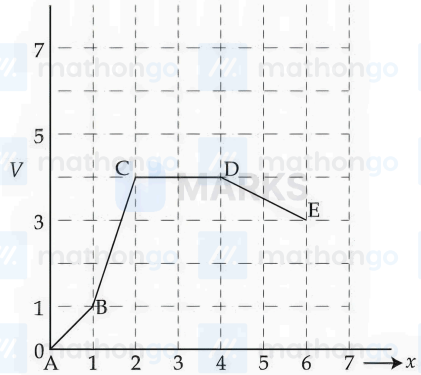
A bead $P$ sliding on a frictionless semi-circular string $(ACB)$ and it is at point $S$ at $t = 0$ and at this instant the horizontal component of its velocity is $v$. Another bead $Q$ of the same mass as $P$ is ejected from point $A$ at $t = 0$ along the horizontal string $AB$, with the speed $v$, friction between the beads and the respective strings may be neglected in both cases. Let $t_P$ and $t_Q$ be the respective times taken by beads $P$ and $Q$ to reach the point $B$, then the relation between $t_P$ and $t_Q$ is




(1) $t_P < t_Q$ (2) $t_P > t_Q$ (3) $t_P = t_Q$ (4) $t_P > 1.25 t_Q$

**Q6. JEE Main 2026 (21 January Shift 1)**

Potential energy ($V$) versus distance ($x$) is given by the graph. Rank various regions as per the magnitudes of the force (F) acting on a particle from high to low.




(1) $F_{CD} > F_{AB} > F_{BC} > F_{DE}$
(2) $F_{CD} > F_{DE} > F_{AB} > F_{BC}$
(3) $F_{BC} > F_{CD} > F_{DE} > F_{AB}$
(4) $F_{BC} > F_{AB} > F_{DE} > F_{CD}$

## ANSWERS AND SOLUTIONS

**1.** (4) **2.** (1) **3.** (1) **4.** (1) **5.** (1) **6.** (4)

**1.** **(4)** Given $x(t) = t^2 + t + 1$, with $m = 2$ kg, find work from $t = 2$ s to $t = 3$ s.

Velocity: $v(t) = \frac{dx}{dt} = 2t + 1$
At $t = 2$ s: $v_i = 2(2) + 1 = 5$ m/s
At $t = 3$ s: $v_f = 2(3) + 1 = 7$ m/s

Using the work-energy theorem:
$W = \Delta KE = \frac{1}{2}m(v_f^2 - v_i^2) = \frac{1}{2}(2)(7^2 - 5^2) = 1(49 - 25) = 24$ J

**2.** **(1)** Bob A swings from $60^\circ$ to vertical. Height drop $h = l(1 - \cos 60^\circ) = 1 \times 0.5 = 0.5$ m.
Velocity of A at bottom: $v_A = \sqrt{2gh} = \sqrt{2 \times 10 \times 0.5} = \sqrt{10}$ m/s.
Elastic collision between equal masses: B gets $v_B = \sqrt{10}$ m/s, A stops.
For B to just complete circular loop: $v_B^2 = 5gR$.
$10 = 5 \times 10 \times R \Rightarrow R = \frac{1}{5}$ m.

**3.** **(1)** At the point where tension $T = 0$ and angle from vertical is $60^\circ$ (i.e., $30^\circ$ from horizontal):
$T + mg\cos 60^\circ = \frac{mV^2}{\ell}$
Since $T = 0$:
$V^2 = \frac{g\ell}{2}$
Using mechanical energy conservation from bottom (A) to the point where $T = 0$:
$\frac{1}{2}mu^2 = mg(\ell + \ell\cos 60^\circ) + \frac{1}{2}mV^2$
$u^2 = 2g\ell\left(1 + \frac{1}{2}\right) + \frac{g\ell}{2} = 3g\ell + \frac{g\ell}{2} = \frac{7g\ell}{2}$
$u = \sqrt{\frac{7g\ell}{2}}$

**4.** **(1)** Statement I:
The work done by a force $\vec{F}$ during a displacement from $\vec{r}_1$ to $\vec{r}_2$ is defined as $W = \int_{r_1}^{r_2} \vec{F} \cdot d\vec{r}$.
The expression given in the statement, $W = -\int_{r_1}^{r_2} \vec{F} \cdot d\vec{r}$, is the definition of the change in potential energy ( $\Delta U = -W$), not the work done by the force itself.
Thus, Statement I is false.

Statement II:
By definition, a conservative force is one for which the work done in moving an object between two points is independent of the path taken.
It only depends on the initial and final positions.
Therefore, the claim that work done changes with the path for a conservative force is false.

Since both statements are false, the correct option is (1).

**5.** **(1)** Horizontal displacement of Q is more than P.

$X_Q > X_P$

Horizontal component of velocity is same

So $t_p = \dfrac{x_p}{v}$

$t_Q = \dfrac{x_Q}{v}$

$t_Q > t_p$

**6.** **(4)** Force is related to potential energy by $F = -\frac{dV}{dx}$.

Magnitude of force $|F| = \left|\frac{dV}{dx}\right|$ = |slope of V-x graph|

From graph coordinates: A(1, 0), B(1.5, 1), C(2, 4), D(4, 4), E(5, 3)

$|F_{AB}| = \frac{1-0}{1.5-1} = 2$

$|F_{BC}| = \frac{4-1}{2-1.5} = 6$

$|F_{CD}| = \frac{4-4}{4-2} = 0$

$|F_{DE}| = \frac{|3-4|}{5-4} = 1$

Ranking: $F_{BC} > F_{AB} > F_{DE} > F_{CD}$